GjPest4CMR: LLMs-Assisted Morphology-Aware Fine-grained Goji Pest Image-Text Retrieval

Anonymous Author(s)

Abstract

Fine-grained agricultural pest image-text retrieval demands robust cross-modal matching capable of handling subtle appearance differences and structured multi-sentence descriptions. Yet, small-scale domain data introduces two key challenges in supervision and representation. First, existing corpora typically pair each image with a single coarse paragraph, providing incomplete supervision that omits life-stage appearances and subtle morphological traits, while high-frequency redundant fields shared across pest categories further cause semantic confounding. Second, prevailing retrieval models rely on global pooled alignment, which averages out low-occupancy but discriminative local cues such as stripe layouts, body segmentation, and wing venation, entangling near-neighbor categories. To address these issues, we propose StruMorAlign, a parameter-efficient cross-attention CLIP framework for fine-grained pest image-text retrieval. We first construct GjPest4CMR, an English goji pest image-text retrieval dataset created via LLMs-assisted re-writing with expert-in-the-loop verification, providing multi-sentence structured descriptions. On the modeling side, we design StruPD (Structured Semantic Deconfounding) to provide multi-granularity structured supervision and strengthen coarse-to-fine consistency learning on morphology-relevant local regions, addressing the first challenge; and introduce GateA^2 (Gated Adaptive Alignment) self-distillation with high-momentum EMA teacher for parameter-efficient, morphology-aware cross-modal alignment, addressing the second challenge. On the GjPest4CMR dataset, StruMorAlign achieves mR 78.72% for both image-to-text and text-to-image retrieval, with I2T R@1=53.11% and T2I R@1=51.53%, outperforming the strongest baseline HarMA by +9.72 mR. Ablation studies confirm the complementary benefits of StruPD and GateA^2, and qualitative analyses including t-SNE visualizations and attention heatmaps further validate the effectiveness of our approach.

CCS Concepts

• **Information systems** → **Image search**; • **Computing methodologies** → *Image representations*; • **Applied computing** → Agriculture.

Keywords

Agricultural pest image-text retrieval, Fine-grained retrieval, CLIP, Parameter-efficient fine-tuning, Cross-attention, Multi-scale attention pooling, Multi-view augmentation, Contrastive learning

ACM Reference Format:

Anonymous Author(s). 2026. GjPest4CMR: LLMs-Assisted Morphology-Aware Fine-grained Goji Pest Image-Text Retrieval. In *Woodstock '18: ACM Symposium on Neural Gaze Detection, June 03–05, 2018, Woodstock, NY*. ACM, New York, NY, USA, 9 pages. https://doi.org/XXXXXXX.XXXXXXX

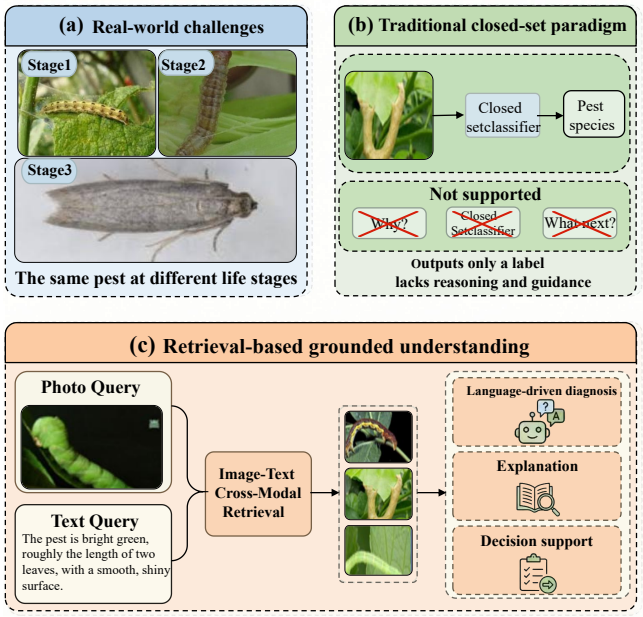


Figure 1: Motivation overview: real-world pest recognition challenges, limitations of closed-set classification, and the proposed retrieval-based grounded understanding paradigm.

1 Introduction

Goji (Wolfberry) is a high-value horticultural crop that plays an important role in regional agriculture and rural economies. Its production, whereas, is highly susceptible to pest outbreaks, where delayed or inaccurate diagnosis can rapidly translate into yield loss and quality degradation. Over the past decade, substantial efforts have been devoted to goji pest understanding, ranging from early image-based recognition systems to modern deep models that automate pest identification under field conditions [14, 18, 27]. Yet, goji orchards remain visually challenging in the wild: *pests appear at different life stages, their morphology changes dramatically across growth phases, and images are frequently captured under cluttered backgrounds, occlusions, and varying illumination*, factors that collectively hinder robust and scalable deployment [14] (see Fig. 1(a)).

In effect, recent goji pest research has begun to shift from vanilla classification/identification toward the multi-modal understanding paradigms (see Fig. 1(b)), such as language-driven diagnosis, visual question answering (VQA), image-text cross-modal retrieval, and interactive decision support [1, 22]. These novel paradigms are appealing because they better match practical workflows: farmers/agronomists do not only ask 'what is it', but also 'why', 'how severe', and 'what to do next'. Unfortunately, enabling such language-centric understanding requires models to connect

visual evidence with structured agronomic knowledge and detailed textual descriptions in a reliable way. Among these paradigms, image-text cross-modal retrieval provides a particularly practical interface for goji pest diagnosis [22]. Instead of forcing users into a closed label space, retrieval supports case-based reasoning: *a practitioner can query with a photo (or a text description) to retrieve similar historical cases, then cross-check pest species, life stage, symptoms, and management suggestions from the retrieved records* (see Fig. 1(c)). In principle, retrieval can serve as a foundation for downstream tasks such as interactive VQA, explanation, and evidence-based decision making, by grounding responses in relevant precedent cases. This raises an essential requirement: the retrieval model must be simultaneously language-general (handle diverse expressions of the same pest), robust to semantic confounding (avoid spurious matches from shared generic phrases), and sensitive to fine-grained morphology (capture subtle local traits for discrimination).

Despite growing interest, current goji pest image-text corpora (e.g., WPIT9K [5]) typically provide Chinese-only annotations, and each image is often paired with a single long paragraph. This design introduces a major bottleneck for real-world deployment [5]. First, monolingual Chinese descriptions mismatch today's vision-language ecosystem, where most pretrained models and prompt interfaces are English-centric. More critically, a single long paragraph is often insufficient to comprehensively describe complex goji pest scenes: it may omit important factors such as background, life-stage-specific appearance, and subtle morphological traits, while mixing multiple aspects into a coarse narrative. As a result, the supervision signal becomes incomplete, weakening fine-grained retrieval reliability (**Challenge 1**; see Fig. 2(a)). Alternatively, goji pest descriptions are commonly structured (e.g., symptom-organ-morphology-season/stage), yet long-form narratives implicitly contain many high-frequency redundant fields shared across pests. When such redundancy dominates the language space, it inevitably causes semantic confusion: query texts may spuriously match non-target images that share common phrases or generic symptom words. Meanwhile, goji pests are distinguished via fine-grained local morphology (e.g., stripe layouts, body segmentation, wing venation, and antenna shapes), whereas prevailing retrieval models tend to rely on global pooled alignment, averaging out low-occupancy but discriminative cues and entangling near-neighbor categories (**Challenge 2**; see Fig. 2(b)). These challenges raise a non-trivial question: *how can we make goji pest cross-modal retrieval language-general, yet semantically deconfounded and morphology-aware?*

In light of the above analysis, we propose StruMorAlign, a parameter-efficient cross-attention CLIP framework that couples structured semantic deconfounding and coarse-to-fine consistency learning with morphology-aware patch-level alignment for fine-grained agricultural pest retrieval. Specifically, we adopt StruPD to provide multi-granularity structured supervision and reduce semantic confounding; at the representation level, we introduce GateA^2 to align image patches with text tokens for capturing morphology-relevant local correspondences (e.g., 'dark stripes', 'segmented body'). To stabilize training under multi-view augmentation, we further employ a high-momentum EMA teacher, and use

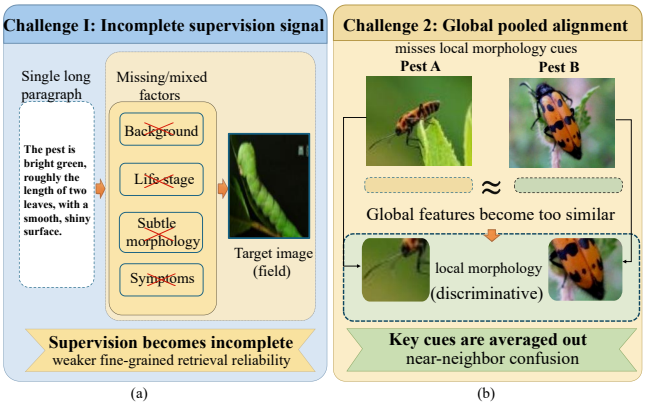


Figure 2: Two core challenges: (a) incomplete supervision signal; (b) global pooled alignment misses local morphology cues.

gated adaptive adapters to achieve parameter-efficient adaptation on small-scale data.

The main contributions of this paper are as follows:

(1) To our knowledge, we are the *first* to construct **GjPest4CMR**, an English goji pest image-text retrieval dataset created via LLMs-assisted re-writing from Chinese annotations, with expert-in-the-loop verification, to reduce hallucinations and subjective bias, realizing semantic coverage of background context and life-stage-dependent appearances.
(2) We propose a novel framework, called StruMorAlign, which integrates **StruPD** for structured semantic deconfounding and coarse-to-fine consistency learning, together with **GateA**^{2} self-distillation for parameter-efficient, morphology-aware cross-modal alignment.
(3) Extensive experiments on goji pest image-text retrieval dataset demonstrate consistent improvements over strong baselines, and the ablations further validate the complementary benefits of StruPD and GateA^2.

The remainder of this paper is organized as follows: Sec. 2 reviews related work; Sec. 3 presents the GjPest4CMR dataset; Sec. 4 describes the StruMorAlign framework; Sec. 5 presents experimental settings and baselines; Sec. 6 reports quantitative results, ablation studies, and qualitative analysis; Sec. 7 concludes the paper.

2 Related Work

This section briefly reviews research progress most relevant to this work, primarily including: (Sec. 2.1) crop pest recognition and detection, (Sec. 2.2) image-text retrieval and cross-modal alignment methods, (Sec. 2.3) vision-language pre-trained models and CLIP adaptation methods.

2.1 Crop Pest Recognition and Detection

Crop pest recognition is a fundamental task in intelligent agriculture, typically formulated as classification, detection, or segmentation [14, 27]. Pest imagery in real farmland environments often exhibits tiny targets, cluttered backgrounds, illumination changes,

and occlusions, making robust recognition challenging [13, 14]. Modern pipelines adopt attention or region-based detectors such as PestNet [18] for multi-class localization, TPest-RCNN [13] for sticky-trap tiny pests, and Pest-YOLO [26] for dense multi-class detection. However, these methods remain closed-set predictors (label/box/mask) and do not support text-driven search, motivating cross-modal retrieval for fine-grained pest scenarios.

2.2 Image-Text Cross-modal Retrieval

Image-text cross-modal retrieval supports bidirectional search (T2I/I2T) by learning aligned visual-text representations. Early work mainly follows dual-encoder embedding frameworks trained with ranking objectives, where hard-negative mining improves retrieval discriminability and scalability [6]. To capture fine-grained correspondences beyond global embeddings, region-word alignment methods introduce cross-attention matching between image regions and words, yielding more accurate similarity estimation [9].

With Transformer-based vision-language pretraining, retrieval models further benefit from stronger multimodal representations: two-stream and single-stream architectures learn cross-modal alignment via objectives such as image-text matching and token/region-level supervision [4, 20]. More recently, large-scale contrastive pretraining has become a dominant backbone. CLIP aligns image/text embeddings on massive image-text pairs and provides efficient dual-encoder retrieval with strong transferability [22]. Subsequent works improve robustness under noisy pairs and enhance fine-grained matching, e.g., ALBEF/BLIP via better pretraining strategies [11, 12] and FILIP via token-level late interaction while retaining indexing efficiency [30]. These advances form the foundation for our domain-specific, fine-grained pest retrieval setting.

2.3 Vision-Language Pre-trained Models

Beyond retrieval, recent multimodal large language models (MLLMs) such as GPT-4 [1] and LLaVA [17] extend vision-language alignment to instruction-following and interactive reasoning. While MLLMs emphasize general understanding, our work focuses on efficient fine-grained retrieval and thus builds upon retrieval-oriented adaptation. Parameter-efficient transfer learning (PETL) methods such as adapter tuning, prompt tuning, and LoRA adapt CLIP to downstream tasks without full fine-tuning [2, 19], substantially reducing overfitting on small-scale datasets. Our GateA^2 builds upon this line by combining gated adapters with cross-attention pooling and self-distillation, specifically tailored for fine-grained morphological alignment.

3 The GjPest4CMR Dataset

To address the limitations of existing goji pest corpora, we construct **GjPest4CMR**, an English goji pest image-text retrieval dataset derived from WPIT9K [5] through LLM-assisted rewriting with expert-in-the-loop verification.

3.1 Dataset Overview

GjPest4CMR contains 9,496 images covering 17 common goji pest species in the Ningxia region of China. The dataset is split into training (7,596), validation (949), and test (951) sets following an 80%/10%/10% ratio. Unlike WPIT9K which provides Chinese-only single-paragraph annotations, GjPest4CMR features structured English descriptions with three complementary sentences per image.

3.2 LLM-Assisted Text Rewriting

Human-written descriptions can be incomplete or subjective, often emphasizing salient symptoms while omitting subtle discriminative cues. To overcome this limitation, we employ GPT-4 to automatically generate structured English descriptions for each image based on the original Chinese annotations [1].

The prompt-based generation method uses a well-designed template with a three-sentence progressive structure:

- **Sentence 1 (Basic Attributes):** Describes fundamental characteristics including pest category, body color, approximate size, and overall shape.
- **Sentence 2 (Structural Posture):** Details morphological structures such as body segmentation, wing venation, antenna shape, leg arrangement, and current posture.
- **Sentence 3 (Background Context):** Captures environmental context including host plant, leaf condition, background color contrast, and life-stage-specific appearance.

This coarse-to-fine annotation strategy ensures comprehensive semantic coverage from macro-level category information to micro-level discriminative details, enabling fine-grained cross-modal alignment.

3.3 Expert-in-the-Loop Verification

To ensure annotation accuracy and reduce hallucinations in LLM-generated content, we implement a two-round expert verification pipeline. In the first round, domain experts in agricultural entomology independently proofread each generated description against the corresponding image, correcting factual errors (e.g., incorrect body color, wrong appendage count, misidentified life stage) and removing hallucinated details not visible in the image. In the second round, a cross-check is performed where a different expert reviews the corrected descriptions to resolve ambiguities and ensure inter-annotator consistency. Before verification, we observed that approximately 18% of GPT-4-generated descriptions contained at least one factual inaccuracy, most commonly involving color attribution and morphological structure confusion between similar species. After the two-round process, the error rate dropped below 3%, confirming the effectiveness of expert-in-the-loop quality control. The final dataset provides high-quality English annotations that better align with mainstream vision-language model interfaces while preserving domain-specific morphological precision.

4 The StruMorAlign Framework

4.1 Overview

As illustrated in Fig. 4, the proposed StruMorAlign framework consists of two core innovation modules that correspond to the two challenges identified in § 1: **StruPD** addresses Challenge 1

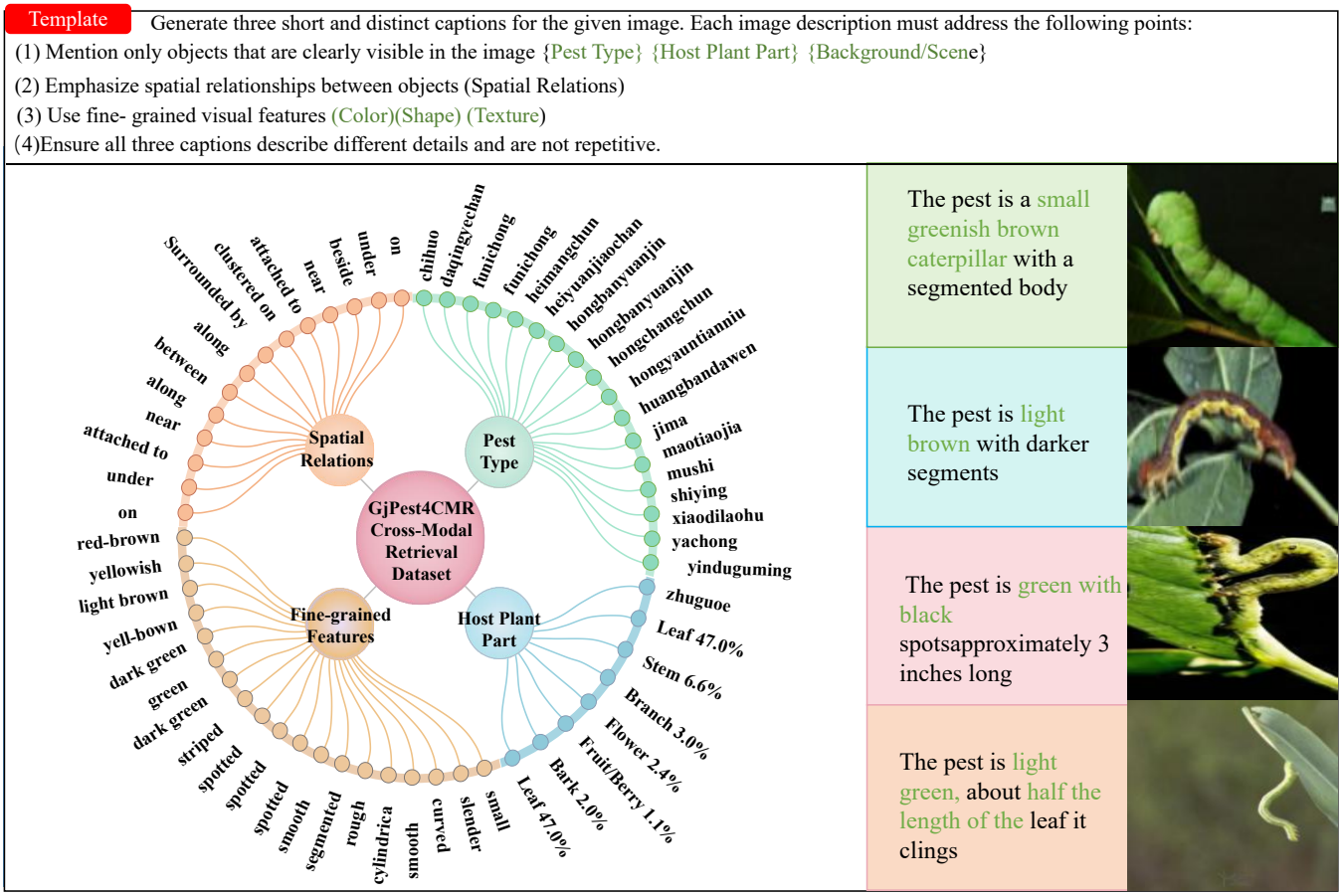


Figure 3: GjPest4CMR dataset schema and example annotations.

by providing structured semantic deconfounding and coarse-to-fine consistency learning; **GateA**^{2} addresses Challenge 2 by enabling parameter-efficient, morphology-aware cross-modal alignment with self-distillation. Inspired by the cross-modality self-distillation paradigm of COSMOS [8], we build upon frozen CLIP encoders and describe each module in detail below.

4.2 StruPD

StruPD provides structured semantic deconfounding via multi-view construction and chunked contrastive learning.

4.2.1 Structured Multi-view Construction. StruPD transforms images into structured view sets consistent with the three-sentence semantic structure (basic attributes → structural posture → background context). For images, we use random cropping to generate $K_{\text{img}} = 4$ views: 2 global views (crop ratio 0.5-1.0) capturing overall appearance, and 2 local views (crop ratio 0.14-0.5) focusing on discriminative details. For texts, we design a cropping strategy aligned with the three-sentence structure, generating $K_{\text{txt}} = 4$ views: 2 global views (randomly combining 1-3 sentences) and 2 local views (single sentence). This structured augmentation ensures that each view pair captures semantically coherent information at matching granularity levels.

4.2.2 Chunked Contrastive Learning. Standard contrastive learning suffers from false negatives when multi-view augmentation is applied: different crops of the same image appear in the same batch and are incorrectly treated as negatives. To address this, StruPD decomposes the multi-view contrastive task into $K_{\text{img}} \times K_{\text{txt}}$ independent blocks.

For each view pair (i, j) where $i \in \{1, \ldots, K_{\text{img}}\}$ and $j \in \{1, \ldots, K_{\text{txt}}\}$, we compute an independent similarity matrix $S^{(i,j)} \in \mathbb{R}^{B\times B}$:

$$S^{(i,j)}_{mn} = s(f^{(i)}_{\text{img}}(I_m), f^{(j)}_{\text{txt}}(T_n)), \quad m, n = 1, \ldots, B \tag{1}$$

where each block contains only image features from view i and text features from view j, ensuring different views of the same image never appear in the same block. The chunked contrastive loss is:

$$\mathcal{L}_{contrast} = \frac{1}{K_{\text{img}} \times K_{\text{txt}}} \sum_{i=1}^{K_{\text{img}}} \sum_{j=1}^{K_{\text{txt}}} \left(\mathcal{L}^{(i,j)}_{i2t} + \mathcal{L}^{(i,j)}_{t2i} \right) \tag{2}$$

This chunking strategy mathematically guarantees gradient consistency, eliminating false negative interference and enabling stable coarse-to-fine consistency learning.

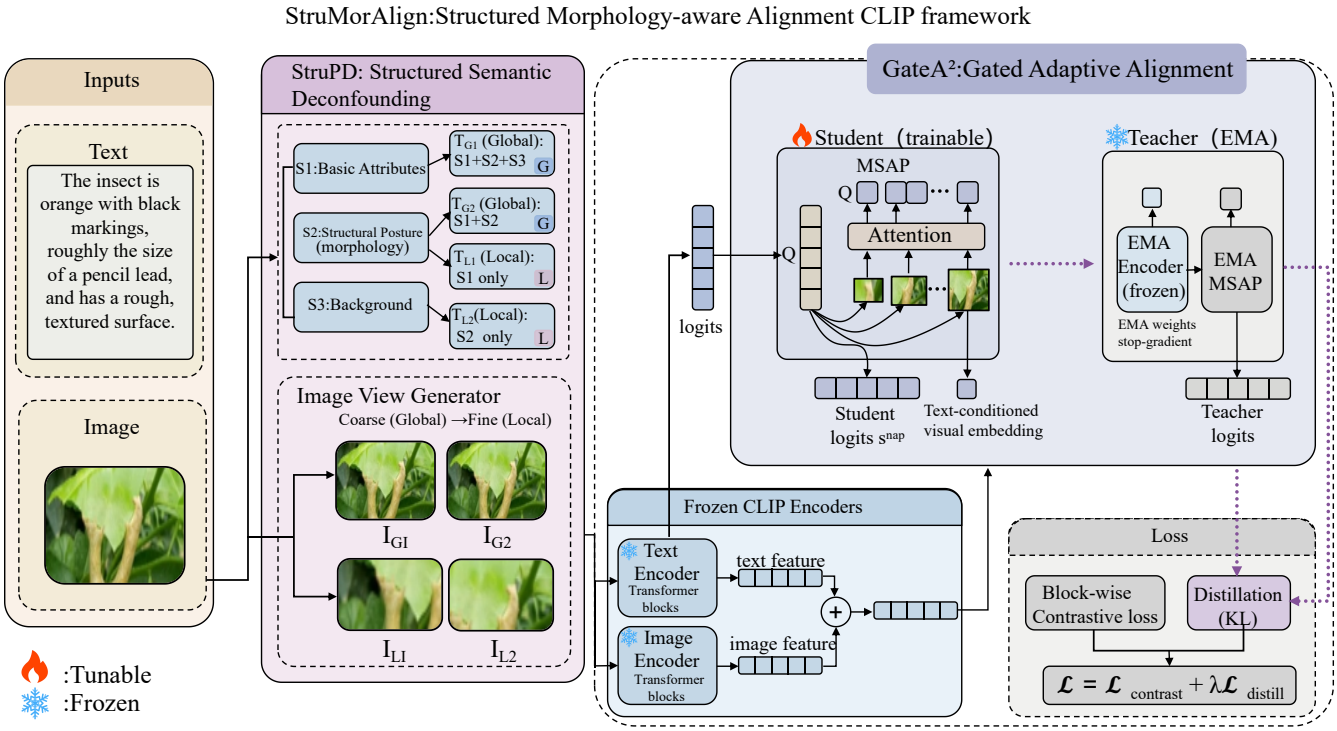


Figure 4: Overall architecture of the StruMorAlign framework. It consists of StruPD for structured semantic deconfounding and GateA^2 for morphology-aware cross-modal alignment.

4.3 GateA^2

GateA^2 is a parameter-efficient alignment module comprising gated adapters, multi-scale attention pooling, and EMA self-distillation.

4.3.1 Gated Adaptive Adapters. To prevent overfitting on small-data scenarios while enabling task-specific adaptation, GateA^2 inserts lightweight adapters into each Transformer layer of ViT, training only adapter parameters while freezing pre-trained weights.

1) Adapter Architecture: A standard adapter consists of down-projection, non-linear activation, and up-projection:

$$h_{\text{adapter}} = W_{\text{up}} \cdot \text{GELU}(W_{\text{down}} \cdot h + b_{\text{down}}) + b_{\text{up}} \tag{3}$$

where $W_{\text{down}} \in \mathbb{R}^{r \times D}$ and $W_{\text{up}} \in \mathbb{R}^{D \times r}$ are down-projection and up-projection matrices, and r is the bottleneck dimension ($r \ll D$). In our implementation, $r = 64$ and $D = 768$, making adapter parameters only 8.3% of the original layer.

2) Gating Mechanism: We introduce a learnable gating coefficient $\alpha \in (0, 1)$ that adaptively mixes pre-trained and adapter outputs:

$$h_{\text{out}} = (1 - \alpha) \cdot h_{\text{pretrain}} + \alpha \cdot h_{\text{adapter}} \tag{4}$$

where $\alpha = \sigma(g)$ with learnable g. This provides implicit regularization (small α preserves pre-trained knowledge) and parameter efficiency (only 3.5M/149.6M = 2.3% trainable parameters).

4.3.2 Multi-Scale Attention Pooling. Standard CLIP uses only the [CLS] token, discarding 196 spatial tokens. GateA^2 leverages all spatial tokens via a two-step pooling mechanism: (1) $L = 4$ learnable queries cross-attend to the full patch sequence to produce L multi-scale summaries; (2) the text EOS embedding re-weights these summaries for the final image representation.

Step 1 (Learnable Cross-Attention): Each learnable query q_l ($l = 1, \ldots, L$) attends to all 196 image patch tokens $H^{\text{img}} = \{h_1^{\text{img}}, \ldots, h_{196}^{\text{img}}\}$:

$$\hat{h}_l = \sum_{i=1}^{196} \alpha_{l,i} \cdot h_i^{\text{img}}, \quad \alpha_{l,i} = \frac{\exp(q_l \cdot (h_i^{\text{img}})^T / \sqrt{D})}{\sum_k \exp(q_l \cdot (h_k^{\text{img}})^T / \sqrt{D})} \tag{5}$$

Step 2 (Text-Guided Aggregation): The text EOS embedding $h_{\text{EOS}}^{\text{txt}}$ re-weights the L summary vectors to yield the final image feature:

$$f_{\text{img}}^{\text{final}} = \sum_{l=1}^{L} \beta_l \cdot \hat{h}_l, \quad \beta_l = \text{softmax}\left(\frac{h_{\text{EOS}}^{\text{txt}} \cdot \hat{h}_l^T}{\sqrt{D}} \right) \tag{6}$$

This two-step design enables dynamic, text-relevant spatial aggregation while preserving multi-scale information for token-level fine-grained alignment.

4.3.3 EMA Self-Distillation. To stabilize training, we employ EMA-based self-distillation. The Teacher model θ_t is updated via $\theta_t \leftarrow m \cdot \theta_t + (1 - m) \cdot \theta_s$ with $m = 0.9995$. The distillation loss

enforces consistency between Student and Teacher predictions:

$$\mathcal{L}_{distill} = \mathrm{KL}\left(P_s(T|I) \| P_t(T|I)\right) + \mathrm{KL}\left(P_s(I|T) \| P_t(I|T)\right) \quad (7)$$

This provides temporal smoothing and denoising effects for stable convergence.

4.4 Overall Loss Function

The final training objective is a weighted combination of block-wise contrastive loss and distillation loss:

$$\mathcal{L} = \mathcal{L}_{contrast} + \lambda \cdot \mathcal{L}_{distill} \quad (8)$$

where λ is the weight coefficient for distillation loss, set to $\lambda = 1.0$ in our experiments. Additionally, we apply L2 regularization ($\lambda_r = 0.01$) to adapter parameters to further prevent overfitting.

During training, we freeze CLIP's pre-trained weights and only optimize adapter parameters, GateA^2 module parameters, and gating coefficients. This ensures parameter efficiency while preserving the powerful representation capability of the pre-trained model.

5 Experiments

5.1 Dataset

The GjPest4CMR dataset is an English goji pest image-text retrieval dataset derived from WPIT9K [5] via LLMs-assisted rewriting with expert verification. It contains 9,496 images covering 17 pest species, split into training (7,596), validation (949), and test (951) sets at 80%/10%/10% ratio. Each image is paired with three English text descriptions emphasizing different morphological features. Image resolution is standardized to 224×224 pixels.

5.2 Evaluation Metrics

We evaluate method performance using R@K (K=1, 5, 10) and mR metrics. The R@K metric assesses the percentage of correctly matched pairs within the top-K ranked results:

$$\mathrm{R@K} = \frac{1}{N} \sum_{i=1}^{N} \mathbb{1}(\mathrm{rank}_i \leq K) \quad (9)$$

where N is the total number of queries and $\mathbb{1}(\cdot)$ is the indicator function. The mR metric represents the mean of all six Recall values across both retrieval directions:

$$\mathrm{mR} = \frac{1}{6} \sum_{d \in \{\mathrm{I2T}, \mathrm{T2I}\}} \sum_{K \in \{1,5,10\}} \mathrm{R@K}^d \quad (10)$$

providing an overall assessment of bidirectional retrieval performance.

5.3 Implementation Details

All experiments are conducted on a single NVIDIA RTX 4090 24GB GPU. Image data augmentation techniques during training include random horizontal flipping, RandomResizedCrop, color jittering, grayscale conversion, Gaussian blur, and solarization. Input images are resized to 224×224 pixel resolution. The maximum text token sequence length is set to 77. The model optimizer uses AdamW with training for 50 epochs. The learning rate is initialized to 5×10^{-4} with cosine annealing scheduling. The first 500 steps constitute the warmup phase. Weight decay is set to 0.1. Batch size is set to 32. For EMA teacher model updates, the momentum parameter is set to 0.9995. Training precision uses mixed precision (AMP) to optimize memory usage. The random seed is fixed at 42 to ensure experimental reproducibility.

5.4 Baselines

To validate the effectiveness of StruMorAlign, we compare it with various existing image-text retrieval methods categorized into two classes: (1) *Traditional methods*: AMFMN [33], GaLR [32], CREAMY [24], GSMN [16], GSC [37], PIR [21]; (2) *CLIP-based methods*: OSA [10], EBAKER [31], IRRA [7], FSRL [34], FMFA [15], DCEL [29], FLAIR [28], DM-adapter [19], PLOT [2], LG-MGC [35], MLM4text [25], AAHR [3], HarMA [36].

6 Experimental Results and Analysis

In this section, we systematically conduct experiments to evaluate the effectiveness of the proposed StruMorAlign framework. We first compare StruMorAlign with state-of-the-art methods, then perform ablation studies to analyze the contribution of each component, and finally present qualitative results for intuitive understanding.

6.1 Comparison with State-of-the-Art Methods

Tab. 1 summarizes the quantitative comparison with prior methods. Our method consistently surpasses existing approaches across all retrieval metrics on GjPest4CMR. For image-to-text retrieval, StruMorAlign achieves R@1=53.11%, while for text-to-image retrieval it reaches R@1=51.53%, yielding an overall mR of 78.72%. Compared with the strongest baselines, StruMorAlign improves mR by +9.72 over HarMA (69.00) and +25.86 over FLAIR (52.86), demonstrating clear advantages on fine-grained pest retrieval. These gains indicate that StruPD effectively leverages structured multi-sentence supervision, while GateA^2 enhances token-level alignment and training stability, together delivering robust bidirectional retrieval performance under small-data conditions.

6.2 Ablation Studies

Tab. 2 summarizes the ablation results. Adding StruPD alone lifts the baseline mR from 53.25 to 54.40, with notable I2T gains (R@1: 28.87→34.77), confirming that structured multi-sentence supervision strengthens coarse-to-fine consistency learning. However, StruPD alone slightly hurts T2I recall, likely because richer text views increase false-negative risk without morphology-level alignment. GateA^2 alone achieves a much larger boost (mR: 53.25→72.53, +19.28), with balanced gains on both I2T and T2I, highlighting the critical role of token-level alignment and EMA-based training stabilization. Combining both modules yields the best performance (mR=78.72, +25.47 over baseline), where T2I R@1 jumps from 45.10 to 51.53 compared with GateA^2 alone, demonstrating clear complementarity: StruPD provides richer multi-granularity supervision signals that GateA^2 can effectively exploit for fine-grained morphological matching.

Table 1: Comparison results of image-text retrieval methods.

Method	Backbone	Image-to-Text			Text-to-Image			mR
		R@1	R@5	R@10	R@1	R@5	R@10	
Traditional methods								
AMFMN [33]	ResNet / LSTM-GRU	5.48	15.34	34.24	6.47	18.45	37.18	19.53
GaLR [32]	ResNet / Seq2Vec	6.47	18.41	36.47	9.42	21.70	40.00	22.08
CREAMY [24]	Faster R-CNN / GRU	2.35	7.89	11.45	4.63	7.41	16.12	8.31
GSMN [16]	ResNet-101 / Bi-GRU	24.00	53.80	67.30	31.60	68.30	80.50	54.25
GSC [37]	ResNet / RNN	15.21	28.45	32.04	15.63	35.14	41.02	27.92
PIR [21]	SwinTransformer-Tiny / BERT	36.17	72.24	83.18	26.99	56.36	68.73	57.28
CLIP-based methods								
OSA [10]	CLIP ViT-B/32	33.26	66.70	77.06	44.78	78.61	88.94	64.89
EBAKER [31]	CLIP ViT-B/16	31.51	65.86	77.81	38.38	73.60	87.70	62.81
IRRA [7]	CLIP ViT-B/16	33.65	66.56	77.60	46.90	78.55	88.01	65.21
FSRL [34]	CLIP ViT-B/16	31.82	64.52	76.40	45.84	81.14	90.31	65.01
FMFA [15]	CLIP ViT-B/16	36.98	68.95	80.72	49.21	81.07	90.54	67.91
DCEL [29]	CLIP ViT-B/16	34.67	66.77	78.41	45.32	80.97	89.59	65.96
FLAIR [28]	CLIP ViT-B/16	22.47	55.13	70.07	30.07	61.72	77.71	52.86
DM-adapter [19]	CLIP ViT / CLIP	28.08	60.15	71.71	39.12	73.82	84.44	59.55
PLOT [2]	CLIP ViT-B/16	36.70	69.93	80.34	46.16	82.02	90.85	67.67
LG-MGC [35]	DALLE2 Diffusion / CLIP	33.30	64.91	76.45	43.74	78.44	87.91	64.13
MLM4text [25]	CLIP ViT-B/16	36.77	67.79	78.62	47.95	80.02	89.27	66.74
AAHR [3]	OpenCLIP-ResNet / BERT	36.70	68.31	78.90	47.21	80.97	90.12	67.04
HarMA [36]	GeoRSCLIP / GeoRSCLIP	35.80	69.50	80.10	51.20	84.50	92.90	69.00
StruMorAlign (Ours)	**CLIP ViT-B/16**	**53.11**	**90.52**	**95.68**	**51.53**	**87.04**	**94.42**	**78.72**

Table 2: Ablation study results for different module combinations.

Method	Image-to-Text			Text-to-Image			*mR*
	R@1	R@5	R@10	R@1	R@5	R@10	
baseline	28.87	60.69	73.34	26.34	58.58	71.65	53.25
+StruPD	34.77	66.06	77.87	23.71	55.63	68.38	54.40
+GateA^2	46.57	82.08	89.98	45.10	82.08	89.35	72.53
Full Model	**53.11**	**90.52**	**95.68**	**51.53**	**87.04**	**94.42**	**78.72**

6.3 Qualitative Results

Fig. 5 shows t-SNE [23] visualizations: the baseline yields loose, overlapping clusters, while the full model (+StruPD+GateA^2) produces compact, well-separated clusters with image-text pairs co-located. Fig. 6 confirms that progressive module stacking shifts attention from background clutter to morphology-relevant regions. Figs. 7 and 8 present bidirectional retrieval examples: as modules are added, correct matches (blue borders) increase steadily, with the full model accurately capturing pest morphology in both I2T and T2I directions. Remaining failures mainly arise from extreme occlusion or visually similar species.

6.4 Hyperparameter Analysis

We conduct hyperparameter ablation on local crops number and sampled captions number (Fig. 9). Results show non-monotonic relationships:

- **Local crops (Fig. 9(a)):** Performance peaks at 2 (53.11%), drops to 50.26% at 4-6, with 0 achieving 51.21%. This suggests global views dominate for pest morphology.
- **Sampled captions (Fig. 9(b)):** 4 captions optimal (53.11%), declining to ~51% at 6-8. This aligns with the dataset's 3-sentence structure.

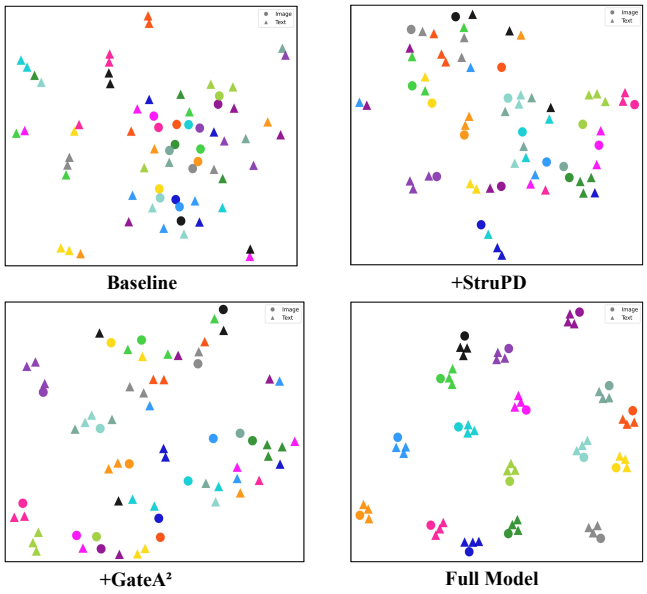


Figure 5: t-SNE visualization results for different model configurations.

Both parameters exhibit clear optima, validating our design choices against the "more is better" assumption.

7 Conclusion

This paper proposes StruMorAlign, a parameter-efficient cross-attention CLIP framework for fine-grained agricultural pest image-text retrieval. To address semantic diversity and morphological alignment challenges, we design StruPD for structured semantic

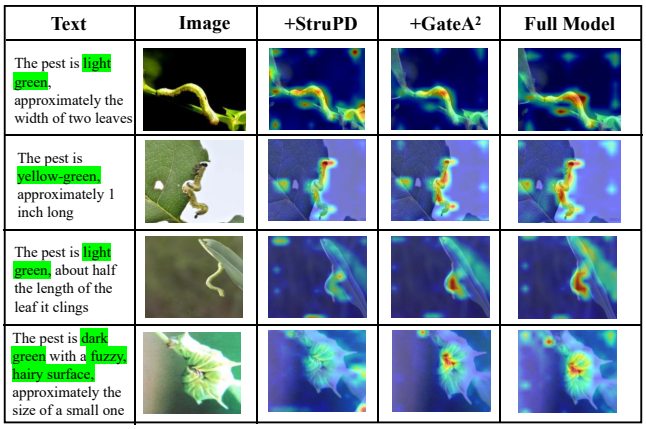


Figure 6: Attention heatmap visualization under different model configurations.

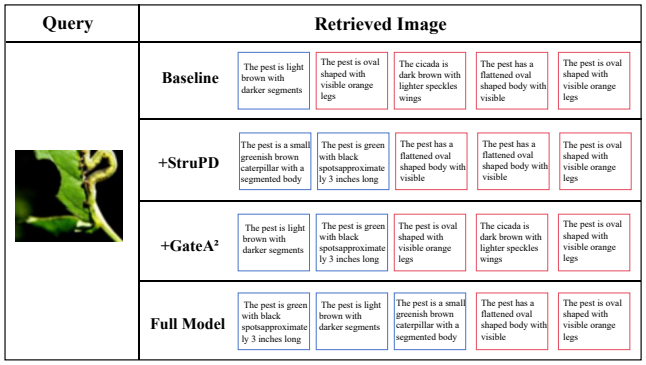


Figure 7: Image-to-text retrieval visualization results.

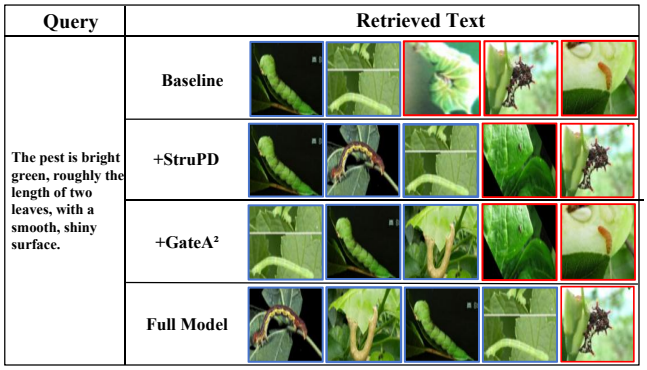
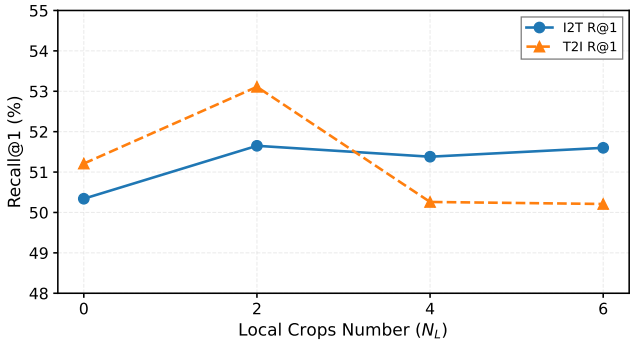
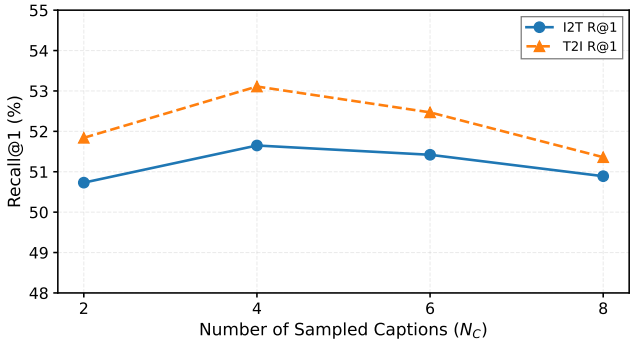


Figure 8: Text-to-image retrieval visualization results.

deconfounding and GateA^2 for morphology-aware cross-modal alignment. On the GjPest4CMR dataset, StruMorAlign achieves mR 78.72% (I2T R@1=53.11%, T2I R@1=51.53%), demonstrating the effectiveness and complementarity of both modules.



(a) Local crops number



(b) Sampled captions number

Figure 9: Hyperparameter ablation studies. (a) Local crops number. (b) Sampled captions number.

References

[1] Joshua Achiam et al. 2023. GPT-4 technical report. *arXiv preprint arXiv:2303.08774* (2023).

[2] Guangyi Chen et al. 2023. PLOT: Prompt learning with optimal transport for vision-language models. In *Proc. of the International Conference on Learning Representations.*

[3] Junyu Chen et al. 2025. Ambiguity-aware and high-order relation learning for multi-grained image-text matching. *Knowledge-Based Systems* (2025).

[4] Yen-Chun Chen, Linjie Li, Licheng Yu, Ahmed El Kholy, Faisal Ahmed, Zhe Gan, Yu Cheng, and Jingjing Liu. 2020. UNITER: Universal image-text representation learning. In *Proc. of the European Conference on Computer Vision (ECCV).*

[5] Guowei Dai, Jingchao Fan, and Christine Dewi. 2023. ITF-WPI: Image and text based cross-modal feature fusion model for wolfberry pest recognition. *Computers and Electronics in Agriculture* 212 (Sept. 2023), 108129. doi:10.1016/j.compag.2023.108129

[6] Fartash Faghri, David J. Fleet, Jamie Ryan Kiros, and Sanja Fidler. 2018. VSE++: Improving visual-semantic embeddings with hard negatives. In *Proc. of the British Machine Vision Conference (BMVC).*

[7] Ding Jiang and Mang Ye. 2023. Cross-modal implicit relation reasoning and alignment for text-based person re-identification. In *Proc. of the IEEE/CVF Conference on Computer Vision and Pattern Recognition.* 2787–2797.

[8] Sanghwan Kim, Rui Xiao, Mariana-Iuliana Georgescu, Stephan Alaniz, and Zeynep Akata. 2025. COSMOS: Cross-modality self-distillation for vision language pre-training. In *Proc. of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR).*

[9] Kuang-Huei Lee, Xi Chen, Gang Hua, Houdong Hu, and Xiaodong He. 2018. Stacked cross attention for image-text matching. In *Proc. of the European Conference on Computer Vision (ECCV).*

[10] Hao Li, Jiayang Gu, Jingkuan Song, An Zhang, and Lianli Gao. 2024. Debiased orthogonal boundary-driven efficient noise mitigation. arXiv:2410.01944 [cs.CV]

[11] Junnan Li, Dongxu Li, Caiming Xiong, and Steven C. H. Hoi. 2022. BLIP: Bootstrapping language-image pre-training for unified vision-language understanding and generation. In *Proc. of the 39th International Conference on Machine*

Learning (ICML).

[12] Junnan Li, Ramprasaath R. Selvaraju, Akhilesh Deepak Gotmare, Shafiq Joty, Caiming Xiong, and Steven C. H. Hoi. 2021. Align before fuse: Vision and language representation learning with momentum distillation. In *Proc. of the Advances in Neural Information Processing Systems (NeurIPS).*

[13] Wenyong Li, Dujin Wang, Ming Li, Yulin Gao, Jianwei Wu, and Xinting Yang. 2021. Field detection of tiny pests from sticky trap images using deep learning in agricultural greenhouse. *Computers and Electronics in Agriculture* 183 (April 2021), 106048. doi:10.1016/j.compag.2021.106048

[14] Wenyong Li, Tengfei Zheng, Zhankui Yang, Ming Li, Chuanheng Sun, and Xinting Yang. 2021. Classification and detection of insects from field images using deep learning for smart pest management: A systematic review. *Ecological Informatics* 66 (December 2021), 101460. doi:10.1016/j.ecoinf.2021.101460

[15] X. Li et al. 2025. Cross-modal full-mode fine-grained alignment for text-to-image person retrieval. *arXiv preprint arXiv:2509.13754* (2025).

[16] Chunxiao Liu, Zhendong Mao, Tianzhu Zhang, Hongtao Xie, Bin Wang, and Yongdong Zhang. 2020. Graph structured network for image-text matching. In *Proc. of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR).*

[17] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. 2023. Visual instruction tuning. *arXiv preprint arXiv:2304.08485* (2023).

[18] Liu Liu, Rujing Wang, Chengjun Xie, Po Yang, Feiyue Wang, Sudirman, and Wencai Liu. 2019. PestNet: An end-to-end deep learning approach for large-scale multi-class pest detection and classification. *IEEE Access* 7 (2019), 45301–45312. doi:10.1109/ACCESS.2019.2909522

[19] Yating Liu et al. 2025. DM-Adapter: Domain-aware mixture-of-adapters for text-based person retrieval. *arXiv preprint arXiv:2503.04144* (2025).

[20] Jiasen Lu, Dhruv Batra, Devi Parikh, and Stefan Lee. 2019. ViLBERT: Pretraining task-agnostic visiolinguistic representations for vision-and-language tasks. In *Proc. of the Advances in Neural Information Processing Systems (NeurIPS).*

[21] Jiancheng Pan, Qing Ma, and Cong Bai. 2023. A prior instruction representation framework for remote sensing image-text retrieval. In *Proc. of the 31st ACM International Conference on Multimedia (MM '23).* 611–620. doi:10.1145/3581783.3612374

[22] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. 2021. Learning transferable visual models from natural language supervision. In *Proc. of the 38th International Conference on Machine Learning (ICML).*

[23] Laurens Van der Maaten and Geoffrey Hinton. 2008. Visualizing data using t-SNE. *Journal of Machine Learning Research* 9, 11 (2008), 2579–2605.

[24] H. Wang et al. 2024. CREAMY: Cross-modal recipe retrieval by avoiding matching imperfectly. *IEEE Transactions on Multimedia* (2024).

[25] L. Wang et al. 2025. Distill CLIP: Enhancing image-text retrieval via cross-modal transformer distillation and masked language modeling. *arXiv preprint arXiv:2505.21549* (2025).

[26] Changji Wen, Hongrui Chen, Zhenyu Ma, Tian Zhang, Ce Yang, Hengqiang Su, and Hongbing Chen. 2022. Pest-YOLO: A model for large-scale multi-class dense and tiny pest detection and counting. *Frontiers in Plant Science* 13 (2022), 973985. doi:10.3389/fpls.2022.973985

[27] Xiaoping Wu, Chi Zhan, Yu-Kun Lai, Ming-Ming Cheng, and Jufeng Yang. 2019. IP102: A large-scale benchmark dataset for insect pest recognition. In *Proc. of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR).*

[28] Rui Xiao, Sanghwan Kim, Mariana-Iuliana Georgescu, Zeynep Akata, and Stephan Alaniz. 2025. FLAIR: VLM with fine-grained language-informed image representations. In *Proc. of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR).* 33533.

[29] Z. Yang et al. 2024. DCEL: Deep cross-modal evidential learning for text-based person retrieval. In *Proc. of the AAAI Conference on Artificial Intelligence.*

[30] Lewei Yao, Runhui Huang, Lu Hou, Guansong Lu, Minzhe Niu, Hang Xu, Xiaodan Liang, Zhenguo Li, Xin Jiang, and Chunjing Xu. 2022. FILIP: Fine-grained interactive language-image pre-training. In *Proc. of the International Conference on Learning Representations (ICLR).*

[31] Zhiqiang Yuan et al. 2024. iEBAKER: Improved remote sensing image-text retrieval framework via eliminate before align and keyword explicit reasoning. *Remote Sensing* 16, 5 (2024).

[32] Zhiqiang Yuan, Wenkai Zhang, et al. 2024. A global-local dual-stream collaborative enhancement model for cross-modal image and text retrieval in remote sensing scenarios. *IEEE Transactions on Geoscience and Remote Sensing* 62 (2024), 1–16.

[33] Zhiqiang Yuan, Wenkai Zhang, Kun Fu, Xuan Li, Chubo Deng, Hongqi Wang, and Xian Sun. 2021. Exploring a fine-grained multiscale method for cross-modal remote sensing image retrieval. *IEEE Transactions on Geoscience and Remote Sensing* (2021). doi:10.1109/TGRS.2021.3078451

[34] B. Zhang et al. 2024. Fine-grained semantics-aware representation learning for text-based person retrieval. *IEEE Transactions on Multimedia* (2024).

[35] Y. Zhang et al. 2024. Overcoming the pitfalls of vision-language model for image-text retrieval. In *Proc. of the International Conference on Learning Representations.*

[36] Zhao Zhang et al. 2024. Efficient remote sensing with harmonized transfer learning and modality alignment. In *Proc. of the International Conference on Learning Representations.*

[37] Zihua Zhao, Mengxi Chen, Tianjie Dai, Jiangchao Yao, Bo Han, Ya Zhang, and Yanfeng Wang. 2024. Mitigating noisy correspondence by geometrical structure consistency learning. In *Proc. of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR).* 27381–27390.